

Statistics Project

Gas Turbine Emissions and AI Maintenance
Analysis

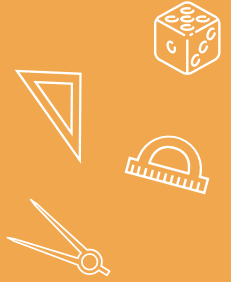




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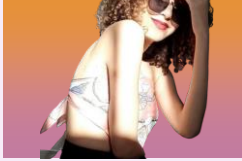
1.

Introduction





Our Group



Donia Ben Othman



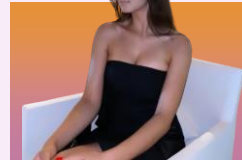
Yosr Lassoued



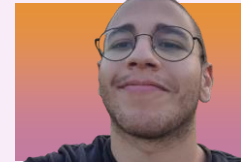
Skander Kechaou



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Arij Mahouechi



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Context

Gas Turbine CO and NOx Emission Data:

- Origin: Collected from a gas turbine located in Turkey.
- Data Characteristics: Includes 11 sensor measures aggregated over one hour, totaling 7,384 observations from 2015.
- Goal: Understand and model relationships between emissions (CO and NOx) and environmental/operational variables (e.g., ambient temperature, pressure, and humidity).

AI4I 2020 Predictive Maintenance Dataset:

- Origin: A synthetic industrial dataset representing predictive maintenance scenarios.
- Data Characteristics: Contains 10,000 observations with variables like product quality variants, rotational speed, torque, and machine failures.
- Goal: Investigate patterns of machine failures and relationships between process parameters and quality metrics.





Problematic

1. Emissions Prediction:

- Assess how environmental and operational factors influence CO and NOx emissions in gas turbines.
- Develop regression models to predict emissions and identify key drivers.

2. Failure Analysis:

- Analyze the relationship between product quality, operational parameters, and machine failure rates.
- Use statistical insights to enhance predictive maintenance and equipment efficiency.

3. Statistical Relationships:

- Investigate correlations among key variables to inform decision-making.
- Validate model assumptions of normality, linearity, and homoscedasticity.

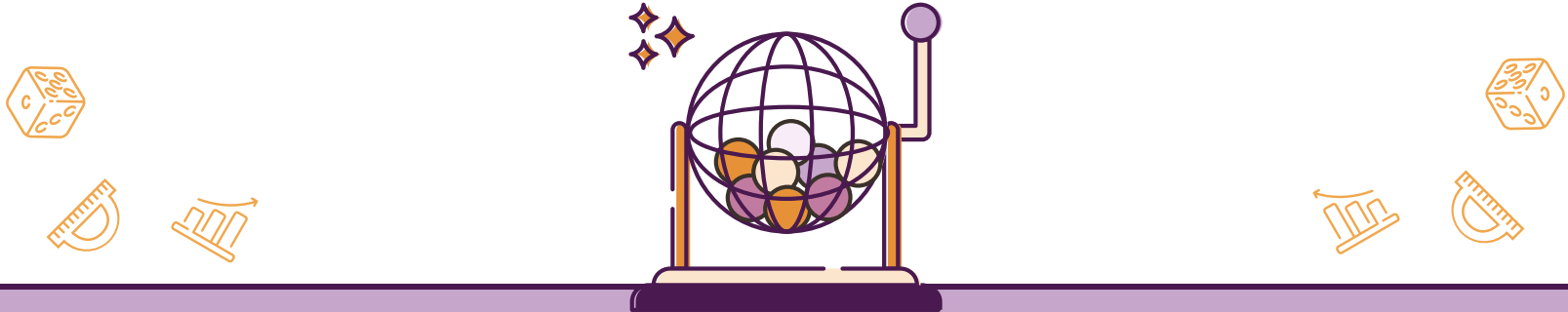
4. Data Optimization:

- Apply statistical methods (ANOVA, T-tests, chi-square) to confirm the significance of patterns.
- Address the impact of outliers and non-linear relationships on model reliability.



2.

Gas Turbine Emissions Analysis



Dataset Overview

Five datasets spanning 2011 to 2015.

Aggregated data of gas turbine operational parameters and emissions.

Key Variables:

CO (Carbon Monoxide): Dependent variable.

NOx (Nitrogen Oxides): Secondary variable of interest.

Independent variables:

- ☐ Ambient Temperature (AT).
- ☐ Ambient Pressure (AP).
- ☐ Ambient Humidity (AH).
- ☐ Air Filter Difference Pressure (AFDP).
- ☐ Gas Turbine Exhaust Pressure (GTEP).
- ☐ Turbine Inlet Temperature (TIT).
- ☐ Turbine After Temperature (TAT).
- ☐ Compressor Discharge Pressure (CDP).
- ☐ Turbine Energy Yield (TEY).



```

> gt_data <- rbind(gt_2011, gt_2012, gt_2013, gt_2014, gt_2015)
> summary(gt_data)
      AT          AP          AH          AFDP
Min.   :-6.235   Min.    : 985.9   Min.    : 24.09   Min.    :2.087
1st Qu.:11.781   1st Qu.:1008.8   1st Qu.: 68.19   1st Qu.:3.356
Median :17.801   Median :1012.6   Median : 80.47   Median :3.938
Mean   :17.713   Mean   :1013.1   Mean   : 77.87   Mean   :3.926
3rd Qu.:23.665   3rd Qu.:1017.0   3rd Qu.: 89.38   3rd Qu.:4.377
Max.   :37.103   Max.   :1036.6   Max.   :100.20   Max.   :7.611

      GTEP        TIT        TAT        TEY
Min.   :17.70   Min.   :1001   Min.   :511.0   Min.   :100.0
1st Qu.:23.13   1st Qu.:1072   1st Qu.:544.7   1st Qu.:124.5
Median :25.10   Median :1086   Median :549.9   Median :133.7
Mean   :25.56   Mean   :1081   Mean   :546.2   Mean   :133.5
3rd Qu.:29.06   3rd Qu.:1097   3rd Qu.:550.0   3rd Qu.:144.1
Max.   :40.72   Max.   :1101   Max.   :550.6   Max.   :179.5

      CDP          CO          NOX
Min.   : 9.852   Min.   : 0.00039   Min.   : 25.91
1st Qu.:11.435   1st Qu.: 1.18240   1st Qu.: 57.16
Median :11.965   Median : 1.71350   Median : 63.85
Mean   :12.061   Mean   : 2.37247   Mean   : 65.29
3rd Qu.:12.855   3rd Qu.: 2.84290   3rd Qu.: 71.55
Max.   :15.159   Max.   :44.10300   Max.   :119.91

> str(gt_data)
'data.frame':   36733 obs. of  11 variables:
 $ AT  : num  4.59 4.29 3.9 3.74 3.75 ...
 $ AP  : num  1019 1018 1018 1018 1018 ...
 $ AH  : num  83.7 84.2 84.9 85.4 85.2 ...
 $ AFDP: num  3.58 3.57 3.58 3.58 3.58 ...
 $ GTEP: num  24 24 24 23.9 23.9 ...
 $ TIT  : num  1086 1086 1086 1086 1086 ...
 $ TAT  : num  550 550 550 550 550 ...
 $ TEY  : num  135 135 135 135 135 ...
 $ CDP  : num  11.9 11.9 12 12 11.9 ...
 $ CO   : num  0.327 0.448 0.451 0.231 0.267 ...
 $ NOX  : num  82 82 82 82 82 ...

```

Data Loading and Preparation

- **Steps:**
 1. Imported datasets for each year (2011–2015) using read.csv.
 2. Combined them into a single dataset using rbind.
 3. Checked for missing values:

```
# Loading gas turbine co and nox emission datasets
gt_2011 <- read.csv(file.choose(), header=T)
gt_2012 <- read.csv(file.choose(), header=T)
gt_2013 <- read.csv(file.choose(), header=T)
gt_2014 <- read.csv(file.choose(), header=T)
gt_2015 <- read.csv(file.choose(), header=T)

# Merging gas turbine co and nox emission datasets in one set
gt_data <- rbind(gt_2011, gt_2012, gt_2013, gt_2014, gt_2015)
summary(gt_data)
str(gt_data)

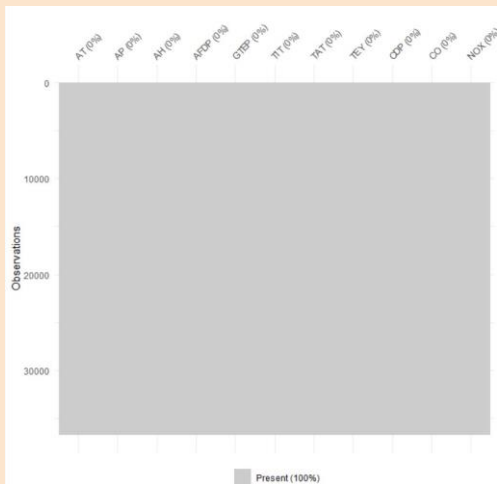
# Check for missing values
# 1) Graphically
vis_miss(gt_data)
# => Graphical representation proves the data is 100% present
# 2) Numerically
mean(is.na(gt_data))
# => The function returns 0 which further proves there is no missing value

# Check for outliers
scaled_gt_data <- scale(gt_data)
boxplot(scaled_gt_data)
# => We scale the data in order to view all outliers on the same scale.
# => The graph proves that every single variable has outliers
```



Check for missing values

Graphically: Used `vis_miss()` for a visual check, confirmed no missing data.



Numerically: Used `mean(is.na(data))`, returning 0, further confirming no missing values.

```
> mean(is.na(gt_data))  
[1] 0
```

Summary Statistics:

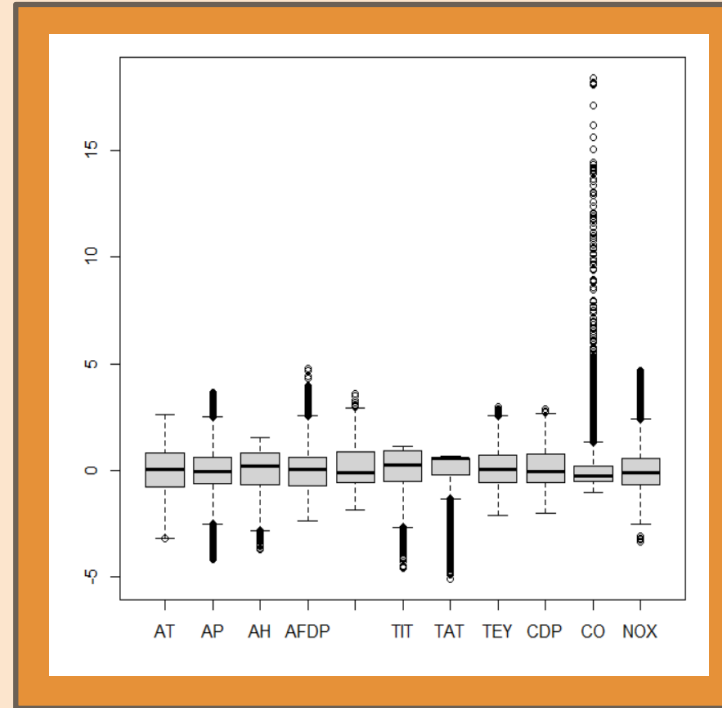
1. Variables had varying ranges, means, and standard deviations.
2. Structure (`str()`) confirmed consistent data types.



Outliers and Imputation

- **Outliers Detected:**
Boxplots of scaled data (`scale()`) showed outliers in nearly every variable.

The graph proves that every single variable has outliers



Imputation Process:

- Developed a function (flag_outliers_na) to replace outliers with NA using the interquartile range (IQR) method:
- Lower bound = $Q1 - 1.5 \times IQR$.
- Upper bound = $Q3 + 1.5 \times IQR$.
- Applied the function across all columns.
- Used KNN (by default 5) imputation to fill NA values, preserving data relationships.

```
# Outlier Imputation
# Function to detect and impute outliers using median to not affect data
flag_outliers_na <- function(column) {
  Q1 <- quantile(column, 0.25)
  Q3 <- quantile(column, 0.75)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR
  column[column < lower_bound | column > upper_bound] <- NA
  return(column)
}

# Apply on all columns
gt_data_na <- as.data.frame(lapply(gt_data, flag_outliers_na))
summary(gt_data_na)

# Impute with KNN
gt_data_prep <- knn(gt_data_na)
summary(gt_data_prep)

# Remove the columns that have '_imp' suffix
gt_data_prep_clean <- gt_data_prep[, !grepl("_imp", colnames(gt_data_prep))]
```



Let's check the cleaned Data

```
> summary(gt_data_prep_clean)
```

AT		AP		AH		AFDP	
Min.	:-6.042	Min.	: 996.5	Min.	: 36.43	Min.	:2.087
1st Qu.:	11.781	1st Qu.:	1008.9	1st Qu.:	68.20	1st Qu.:	3.355
Median :	17.801	Median :	1012.6	Median :	80.47	Median :	3.936
Mean :	17.713	Mean :	1013.1	Mean :	77.92	Mean :	3.904
3rd Qu.:	23.665	3rd Qu.:	1016.9	3rd Qu.:	89.38	3rd Qu.:	4.370
Max.	:37.103	Max.	:1029.3	Max.	:100.20	Max.	:5.909

GTEP		TIT		TAT		TEY	
Min.	:17.70	Min.	:1034	Min.	:536.8	Min.	:100.0
1st Qu.:	23.13	1st Qu.:	1072	1st Qu.:	544.8	1st Qu.:	124.5
Median :	25.10	Median :	1086	Median :	549.9	Median :	133.7
Mean :	25.56	Mean :	1082	Mean :	547.1	Mean :	133.5
3rd Qu.:	29.06	3rd Qu.:	1097	3rd Qu.:	550.0	3rd Qu.:	144.1
Max.	:37.88	Max.	:1101	Max.	:550.6	Max.	:173.5

CDP		CO		NOX	
Min.	: 9.852	Min.	:0.000388	Min.	:35.60
1st Qu.:	11.435	1st Qu.:	1.170100	1st Qu.:	57.16
Median :	11.965	Median :	1.688100	Median :	63.78
Mean :	12.060	Mean :	2.052233	Mean :	64.74
3rd Qu.:	12.855	3rd Qu.:	2.692200	3rd Qu.:	71.37
Max.	:14.976	Max.	:5.333500	Max.	:93.11

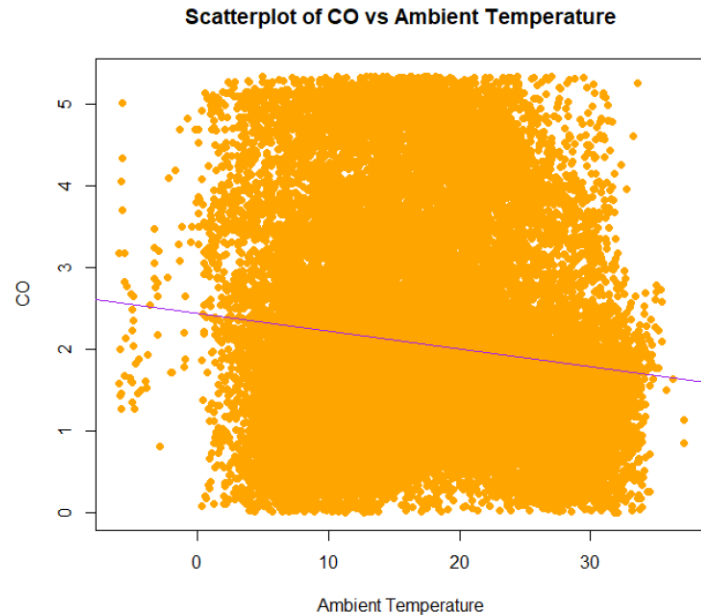


Correlation Analysis



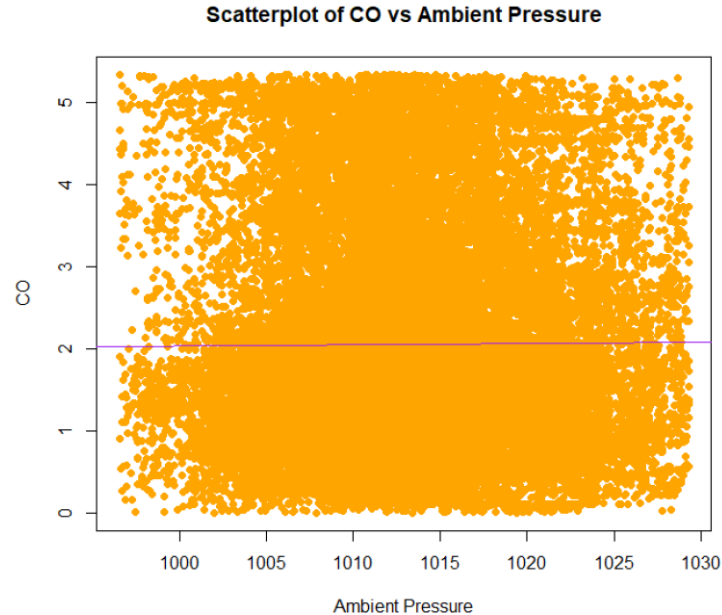
We will check relationships and correlation between variables with our cleaned data in order to understand the linearity

Check relation between independant variables and target CO graphically CO and Ambient Temperature



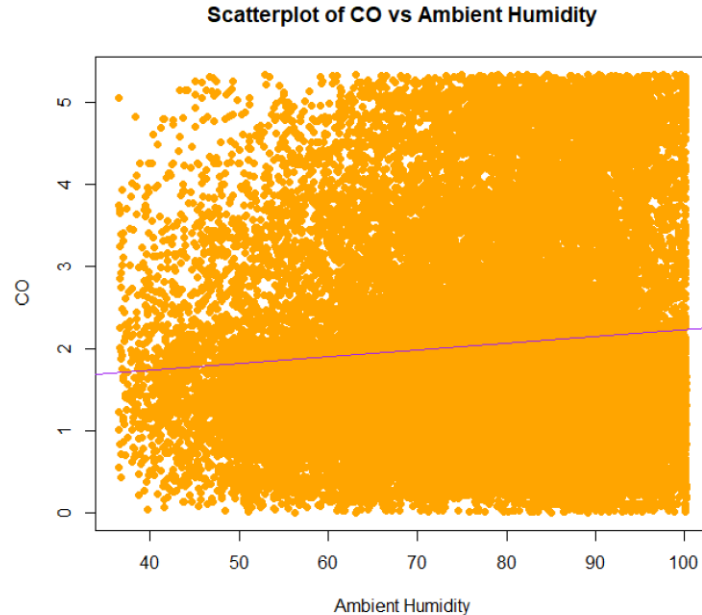
**We can tell CO and AT
aren't related because the
points are loosely
scattered**

Check relation between independant variables and target CO graphically CO and Ambient Pressure



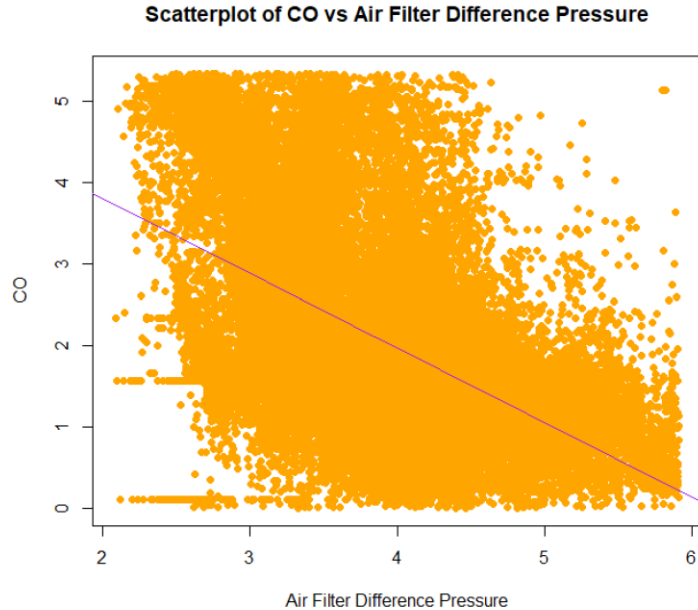
**We can tell CO and AP
aren't related because the
points are loosely
scattered**

Check relation between independant variables and target CO graphically CO and Ambient Humidity



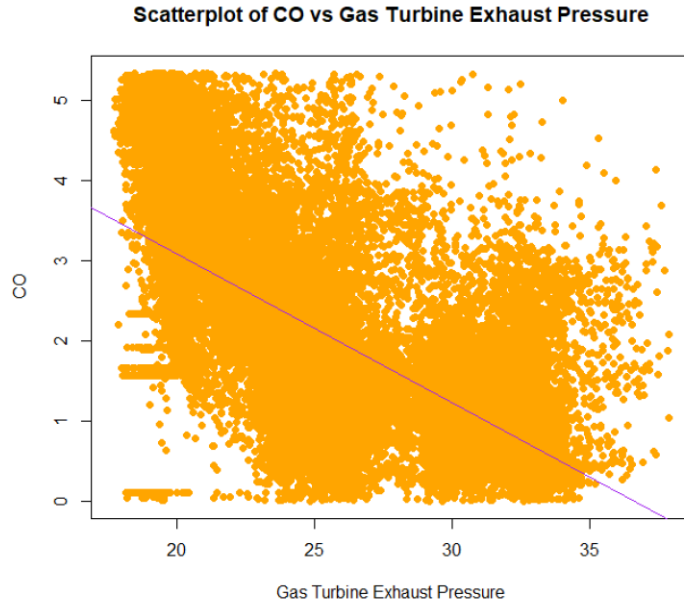
**We can tell CO and AH
aren't related because the
points are loosely
scattered**

Check relation between independant variables and target CO graphically CO and Air Filter Difference Pressure



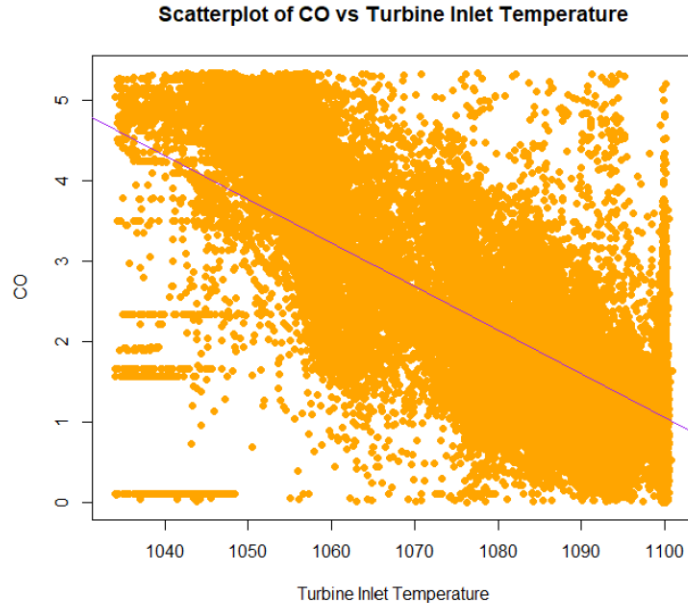
As the x-axis variable increases, the y-axis variable decreases. We can tell CO and AFDP have a negative relationship

Check relation between independant variables and target CO graphically CO and Turbine Exhaust Pressure



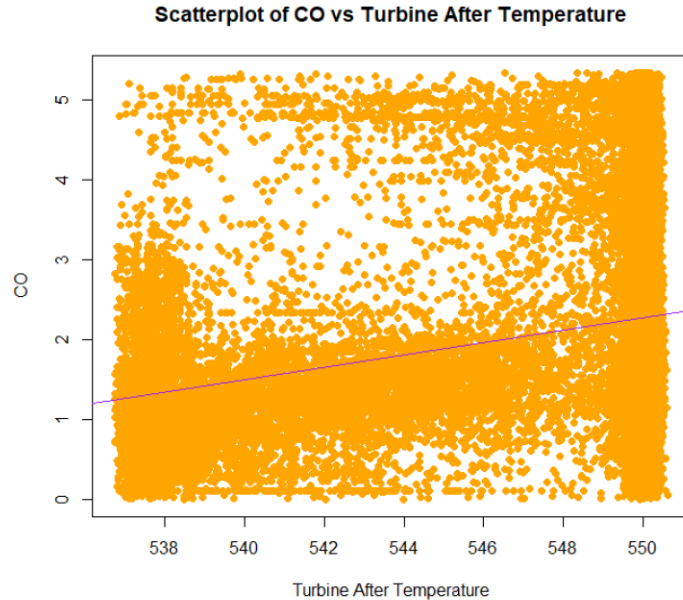
As the x-axis variable increases, the y-axis variable decreases. We can tell CO and GTEP have a negative relationship

Check relation between independant variables and target CO graphically CO and Turbine Inlet Temperature



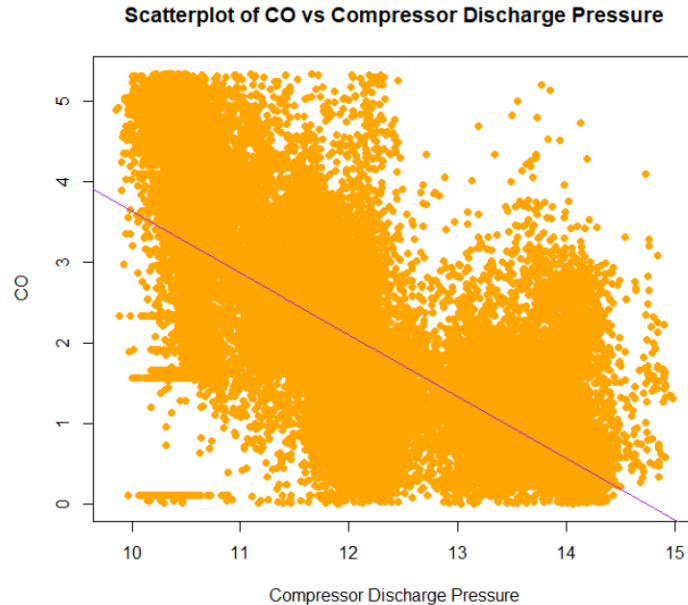
As the x-axis variable increases, the y-axis variable decreases. We can tell CO and TIT have a negative relationship

Check relation between independant variables and target CO graphically CO and Turbine After Temperature



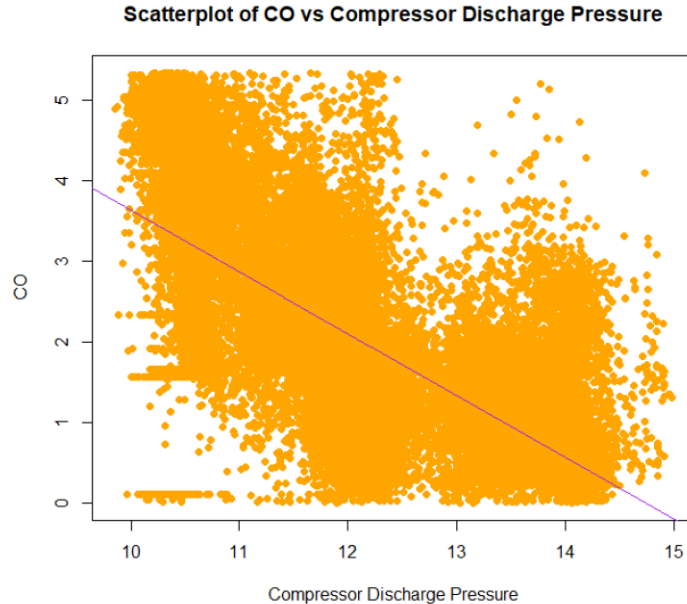
**We can tell CO and TAT
aren't related because the
points are loosely
scattered**

Check relation between independant variables and target CO graphically CO and Compressor Discharge Pressure



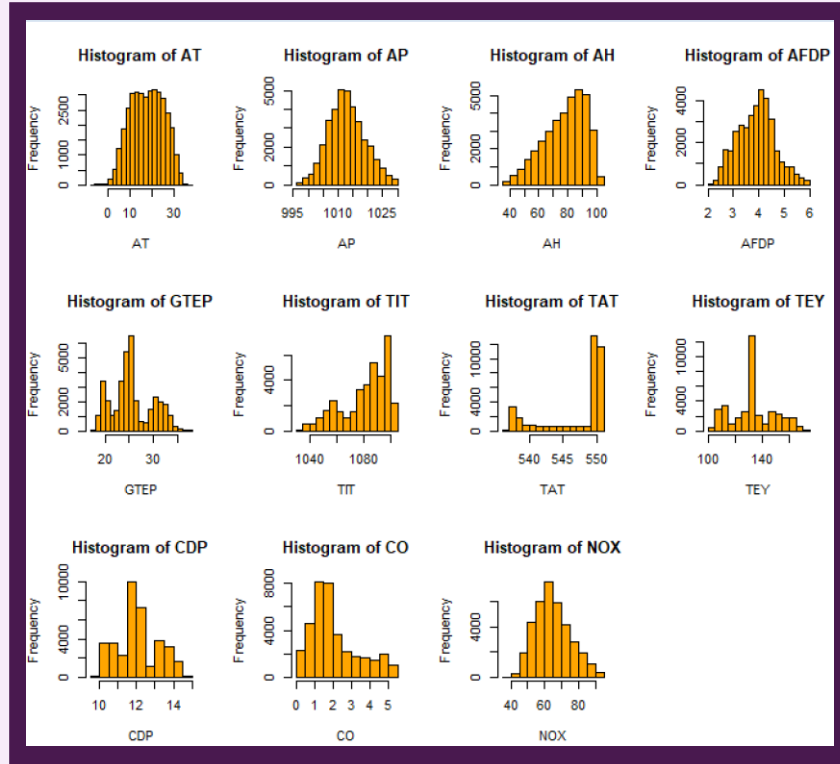
As the x-axis variable increases, the y-axis variable decreases. We can tell CO and CDP have a negative relationship

Check relation between independant variables and target CO graphically CO and Compressor Discharge Pressure



As the x-axis variable increases, the y-axis variable decreases. We can tell CO and CDP have a negative relationship

Variable normality test



**We can remark
that all of our
variables don't
follow a normal
distribution**

Skewness and Kurtosis tests

```
> skewness_values
      AT      AP      AH      AFDP      GTEP      TIT      TAT
-0.04342253  0.18934480 -0.58588217  0.14857780  0.32531844 -0.81418185 -1.23486960
      TEY      CDP      CO      NOX
0.11230296  0.23545491  0.91220792  0.38274435

> kurtosis_values
      AT      AP      AH      AFDP      GTEP      TIT      TAT
-0.82721182 -0.20376744 -0.41545203 -0.27573421 -0.66580264 -0.36752879 -0.18649664
      TEY      CDP      CO      NOX
-0.51129379 -0.63539005 -0.05693873 -0.31536569
```

–Most of our variables are close to symmetric (~ 0) or have mild skewness except TAT

–Most of our variables have low kurtosis < 3 except TAT who's kurtosis > 4



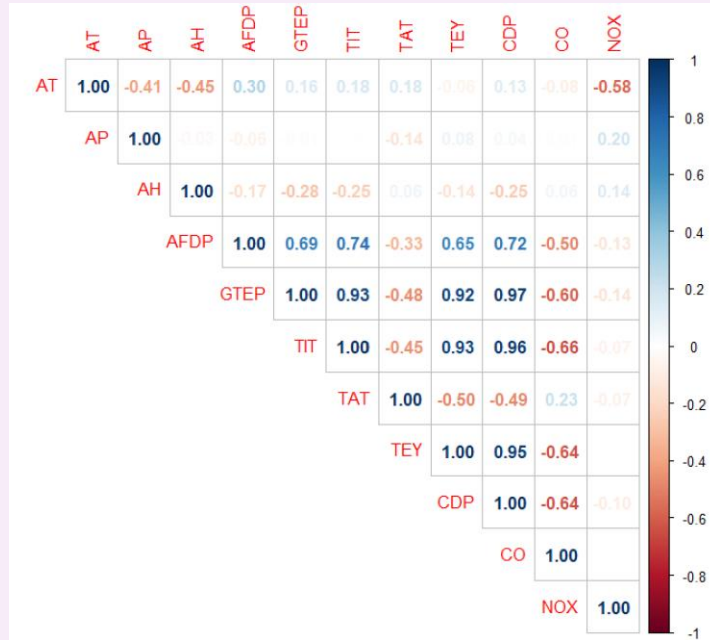
Skewness and Kurtosis tests

Variable	P_Value	Normality
AT	1.232893e-51	Not Normal
AP	6.172766e-35	Not Normal
AH	6.699824e-172	Not Normal
AFDP	1.183774e-26	Not Normal
GTEP	0.000000e+00	Not Normal
TIT	0.000000e+00	Not Normal
TAT	0.000000e+00	Not Normal
TEY	0.000000e+00	Not Normal
CDP	0.000000e+00	Not Normal
CO	0.000000e+00	Not Normal
NOX	1.309626e-64	Not Normal

In conclusion, given the very small p-values (all $< 2.2e-16$), none of the variables in our dataset follow a normal distribution.



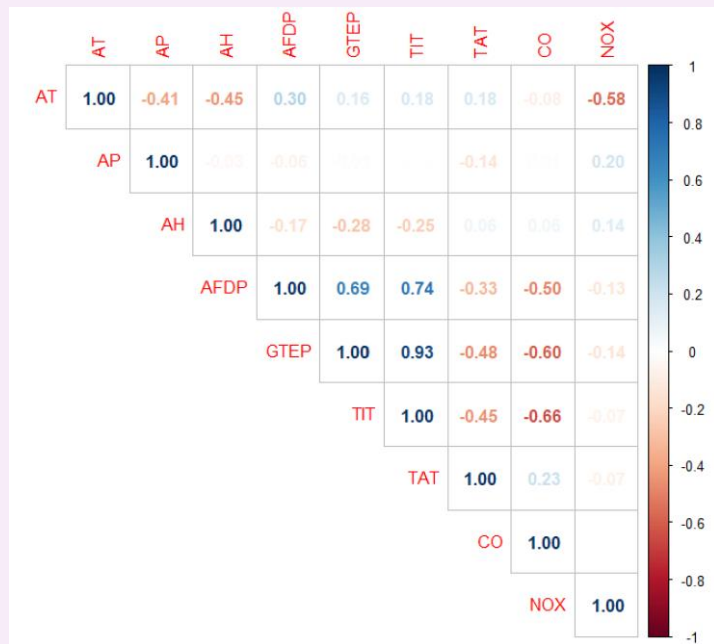
We will use Spearman instead of Pearson because it doesn't need normality.



As a matter of fact, TIT has the strongest relationship with CO (-0.59) followed by CDP, TEY, GTEP and AFDP, whereas NOX only has a strong relationship with AT (-0.6) meaning as temperature decreases, NOX increases. However, GTEP, TIT, TEY and CDP have strong relationships with each other. They provide the same information and we need to remove some variables.



We will use Spearman instead of Pearson because it doesn't need normality.



We remove CDP and TEY because they have the strongest multicollinearity problem.

Regression Analysis

```
Call:
lm(formula = CO ~ AT + AP + AH + AFDP + GTEP + TIT + TAT + NOX,
    data = standardized_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.2169	-0.3841	-0.0359	0.3822	3.2917

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.074e-15	3.506e-03	0.000	1.000000
AT	2.316e-02	6.327e-03	3.660	0.000253 ***
AP	5.000e-03	4.074e-03	1.227	0.219658
AH	-5.342e-02	4.587e-03	-11.645	< 2e-16 ***
AFDP	-1.822e-02	5.415e-03	-3.365	0.000765 ***
GTEP	3.807e-01	1.374e-02	27.707	< 2e-16 ***
TIT	-1.007e+00	1.087e-02	-92.644	< 2e-16 ***
TAT	1.320e-01	6.921e-03	19.076	< 2e-16 ***
NOX	5.286e-02	4.863e-03	10.869	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.672 on 36724 degrees of freedom
Multiple R-squared: 0.5485, Adjusted R-squared: 0.5484
F-statistic: 5576 on 8 and 36724 DF, p-value: < 2.2e-16

**We will normalize
our dataset
(using Z-score)
Then render the
Full model with all
predictors**



Perform stepwise regression

Start: AIC=-29188.74

CO ~ AT + AP + AH + AFDP + GTEP + TIT + TAT + NOX

	Df	Sum of Sq	RSS	AIC
- AP	1	0.7	16587	-29189
<none>			16586	-29189
- AFDP	1	5.1	16591	-29179
- AT	1	6.0	16592	-29177
- NOX	1	53.4	16640	-29073
- AH	1	61.2	16647	-29055
- TAT	1	164.4	16751	-28829
- GTEP	1	346.7	16933	-28431
- TIT	1	3876.4	20463	-21476

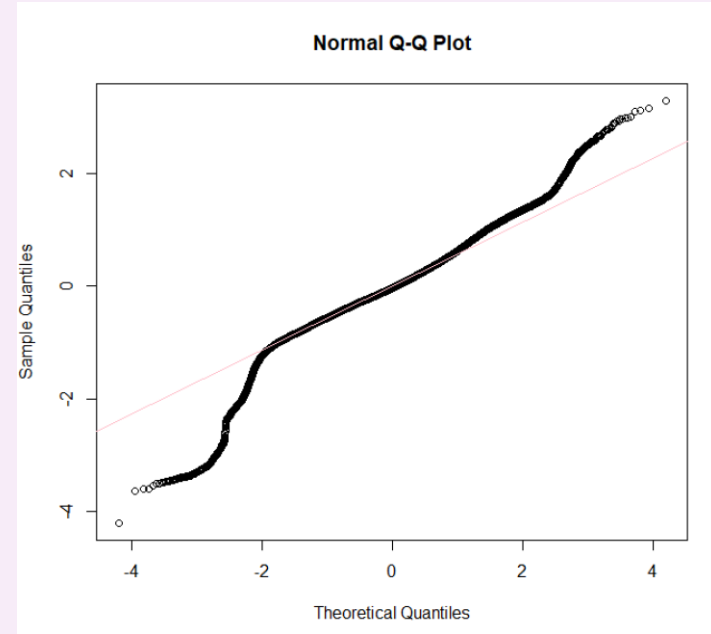
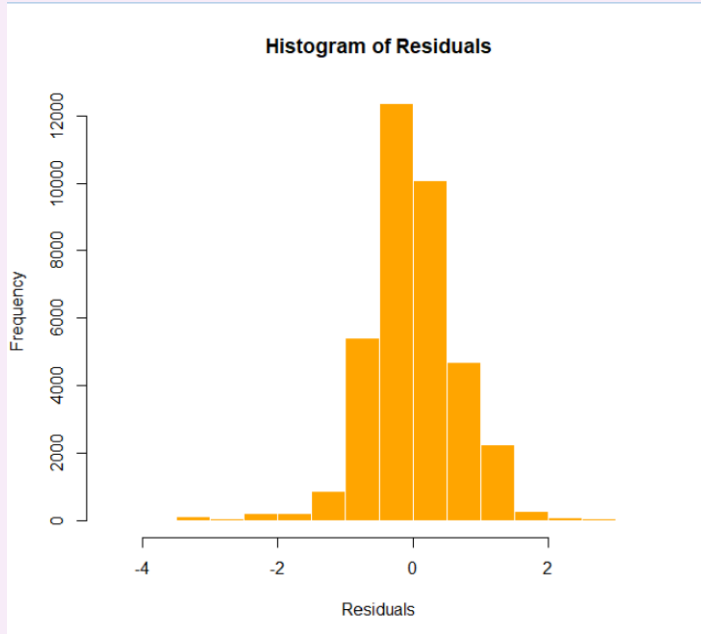
Step: AIC=-29189.23

CO ~ AT + AH + AFDP + GTEP + TIT + TAT + NOX

	Df	Sum of Sq	RSS	AIC
<none>			16587	-29189
+ AP	1	0.7	16586	-29189
- AFDP	1	4.9	16592	-29180
- AT	1	5.5	16592	-29179
- NOX	1	52.8	16640	-29075
- AH	1	71.7	16659	-29033
- TAT	1	164.8	16752	-28828
- GTEP	1	351.7	16939	-28421
- TIT	1	3933.7	20521	-21374

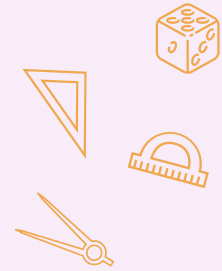
We selected the most statistically significant variables while removing those that do not contribute much to the predictive power.
AP was removed

Residuals



The residuals seem to not follow a normal distribution

Skewness and Kurtosis



Skewness = -0.3155165

**it's close to 0
meaning it's
symmetric**

Kurtosis = 2.849498

**it's less than 3
meaning it's
lightly tailed**



R-squared

R-squared = 0.5484 which means our model explains ~54% of the variance

```
Start: AIC=25236.93
gt_data$CO ~ AT + AP + AH + AFDP + GTEP + TIT + TAT + TEY + CDP + NOX
```

	Df	Sum of Sq	RSS	AIC
<none>			72975	25237
- TAT	1	25.1	73000	25248
- AP	1	116.2	73091	25293
- AH	1	146.4	73121	25309
- TEY	1	173.3	73148	25322
- GTEP	1	232.4	73207	25352
- AFDP	1	291.4	73266	25381
- AT	1	638.7	73613	25555
- CDP	1	741.3	73716	25606
- TIT	1	5888.4	78863	28085
- NOX	1	9124.5	82099	29563

```
Call:
lm(formula = gt_data$CO ~ AT + AP + AH + AFDP + GTEP + TIT + TAT + TEY + CDP + NOX, data = gt_data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-14.241  -0.636  -0.087   0.504  34.409
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.321e+02  2.083e+00  63.434 < 2e-16 ***
AT           5.842e-02  3.258e-03  17.928 < 2e-16 ***
AP           1.082e-02  1.415e-03   7.647 2.11e-14 ***
AH           5.710e-03  6.653e-04   8.582 < 2e-16 ***
AFDP        -1.817e-01  1.500e-02 -12.108 < 2e-16 ***
GTEP         1.036e-01  9.582e-03  10.815 < 2e-16 ***
TIT         -1.577e-01  2.897e-03 -54.434 < 2e-16 ***
TAT          1.319e-02  3.708e-03   3.556 0.000377 ***
TEY         -7.217e-02  7.729e-03  -9.338 < 2e-16 ***
CDP          2.060e+00  1.066e-01  19.314 < 2e-16 ***
NOX          6.146e-02  9.070e-04  67.761 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.41 on 36722 degrees of freedom
Multiple R-squared:  0.612,    Adjusted R-squared:  0.6118
F-statistic: 5791 on 10 and 36722 DF, p-value: < 2.2e-16
```

The R-squared is considerably higher

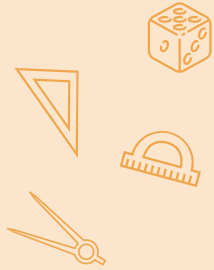
This doesn't necessarily mean the model with outliers explained ~57% of CO variance since the extreme outliers impacted the results.



Key Findings

In a nutshell, the decrease of certain factors (AFDP, GTEP, TIT, TAT) are key drivers of increased CO emissions in gas turbines.

NOX emissions increase the lower Ambient Temperature is. Although the model only explains ~42% of the variance in CO emissions, further research could explore additional factors or refine the model to improve predictive accuracy. Outlier imputation was critical in obtaining more stable and realistic results.



3.

AI4I Analysis



Dataset Overview

- ❑ **Name:** AI4I 2020 Predictive Maintenance Dataset
- ❑ **Source:** Predictive maintenance for industrial machinery
- ❑ **Records:** 10,000 entries
- ❑ **Data Type:** Mixed (Numerical and Categorical)
- ❑ **Purpose:** Analyze factors influencing machine failures and identify potential predictors for maintenance issues.



Variable	Description	Type
UDI	Unique identifier for each record	Categorical
Product_ID	Identifier for the product	Categorical
Type	Product type (Low, Medium, High)	Categorical
Air_Temperature	Air temperature (°C)	Numerical
Process_Temperature	Process temperature (°C)	Numerical
Rotational_Speed	Rotational speed (RPM)	Numerical
Torque	Torque (Nm) 	Numerical
Tool_Wear	Tool wear (minutes)	Numerical
Machine_Failure	Whether the machine failed (0 = No, 1 = Yes)	Binary
TWF	Tool wear failure (0 = No, 1 = Yes)	Binary
HDF	Heat dissipation failure (0 = No, 1 = Yes)	Binary
PWF	Power failure (0 = No, 1 = Yes)	Binary
OSF	Overstrain failure (0 = No, 1 = Yes)	Binary
RNF	Random failure (0 = No, 1 = Yes)	Binary



Data Loading and Preparation

Steps Taken:

1. Loaded the Dataset:

- Used `read.csv()` to load the AI4I 2020 Predictive Maintenance dataset.

2. Renamed Columns for Clarity:

- Simplified column names for better readability.

```
$ UDI           : int  1 2 3 4 5 6 7 8 9 10 ...  
$ Product_ID    : chr  "M14860" "L47181" "L47182" "L47183" ...  
$ Type          : chr  "M" "L" "L" "L" ...  
$ Air_Temperature : num  298 298 298 298 298 ...  
$ Process_Temperature: num  309 309 308 309 309 ...  
$ Rotational_Speed : int  1551 1408 1498 1433 1408 1425 1558 1527 1667 1741  
$ Torque        : num  42.8 46.3 49.4 39.5 40 41.9 42.4 40.2 28.6 28 ...  
$ Tool_Wear      : int   0 3 5 7 9 11 14 16 18 21 ...  
$ Machine_Failure : int   0 0 0 0 0 0 0 0 0 0 ...  
$ TWF           : int   0 0 0 0 0 0 0 0 0 0 ...  
$ HDF           : int   0 0 0 0 0 0 0 0 0 0 ...  
$ PWF           : int   0 0 0 0 0 0 0 0 0 0 ...  
$ OSF           : int   0 0 0 0 0 0 0 0 0 0 ...  
$ RNF           : int   0 0 0 0 0 0 0 0 0 0 ...
```



Data Loading and Preparation

Steps Taken:

3. Checked for Missing Values:

- **Graphical Check:** Used `vis_miss()` to visualize missing data.
- **Numerical Check:** Used `mean(is.na(ai4i2020))` to verify the absence of missing values.



Outliers and Imputation

Steps Taken:

2. Outlier Imputation with

`flag_outliers_na()`

Description:

- **Action:**
 - Outliers were replaced with NA using a custom function that calculates the lower and upper bounds based on the Interquartile Range (IQR).
- **Result:**
 - The summary statistics after this step show NA values in place of outliers for **Rotational Speed** and **Torque**.



```
> summary(ai4i_na)
```

UDI	Product_ID	Type	Air_Temperature
Min. : 1	Length:10000	Length:10000	Min. :295.3
1st Qu.: 2501	Class :character	Class :character	1st Qu.:298.3
Median : 5000	Mode :character	Mode :character	Median :300.1
Mean : 5000			Mean :300.0
3rd Qu.: 7500			3rd Qu.:301.5
Max. :10000			Max. :304.5
Process_Temperature	Rotational_Speed	Torque	Tool_Wear
Min. :305.7	Min. :1168	Min. : 3.80	Min. : 0
1st Qu.:308.8	1st Qu.:1423	1st Qu.:33.20	1st Qu.: 53
Median :310.1	Median :1503	Median :40.10	Median :108
Mean :310.0	Mean :1539	Mean :39.99	Mean :108
3rd Qu.:311.1	3rd Qu.:1612	3rd Qu.:46.80	3rd Qu.:162
Max. :313.8	Max. :2886	Max. :76.60	Max. :253
Machine_Failure	TWF	HDF	PWF
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.0339	Mean :0.0046	Mean :0.0115	Mean :0.0095
3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.0000
OSF	RNF		
Min. :0.0000	Min. :0.0000		
1st Qu.:0.0000	1st Qu.:0.0000		
Median :0.0000	Median :0.0000		
Mean :0.0098	Mean :0.0019		
3rd Qu.:0.0000	3rd Qu.:0.0000		
Max. :1.0000	Max. :1.0000		



Outliers and Imputation

Steps Taken:

3. kNN Imputation Results

Description:

- **Action:**
 - The `kNN()` method was applied to impute the missing values (NA) caused by outlier replacement.
- **Result:**
 - The summary statistics now show that the NA values have been filled with appropriate values derived from the nearest neighbors.

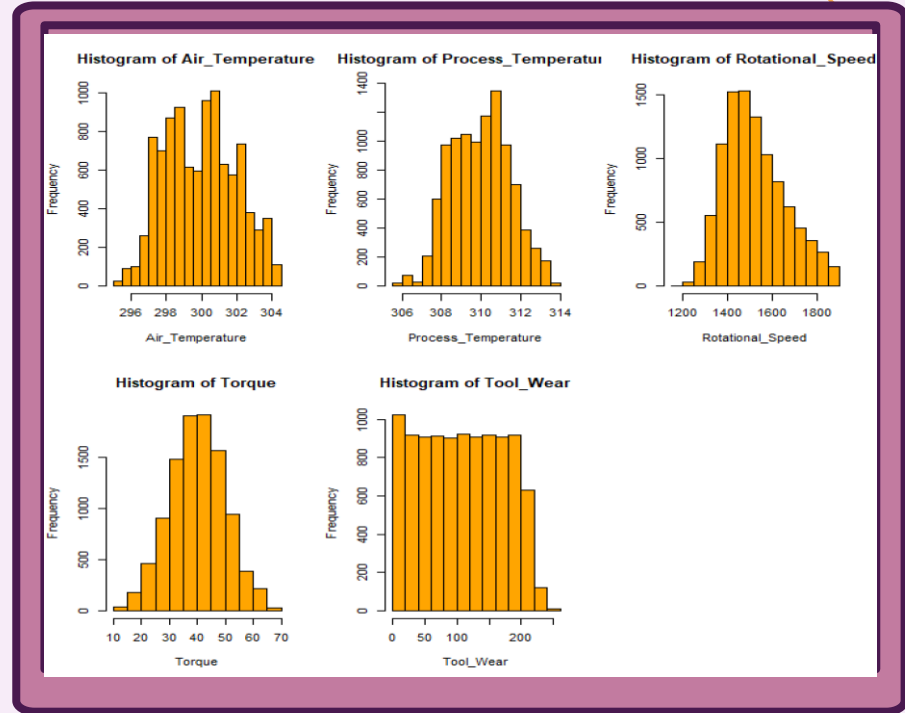
```
> summary(ai4i_na)
      UDI      Product_ID      Type      Air_Temperature
Min.   : 1      Length:10000      Length:10000      Min.   :295.3
1st Qu.:2501      Class :character      Class :character      1st Qu.:298.3
Median :5000      Mode  :character      Mode  :character      Median :300.1
Mean   :5000
3rd Qu.:7500
Max.   :10000
Process_Temperature Rotational_Speed Torque Tool_Wear
Min.   :305.7      Min.   :1168      Min.   : 3.80      Min.   : 0
1st Qu.:308.8      1st Qu.:1423      1st Qu.:33.20      1st Qu.: 53
Median :310.1      Median :1503      Median :40.10      Median :108
Mean   :310.0      Mean   :1539      Mean   :39.99      Mean   :108
3rd Qu.:311.1      3rd Qu.:1612      3rd Qu.:46.80      3rd Qu.:162
Max.   :313.8      Max.   :2886      Max.   :76.60      Max.   :253
Machine_Failure      TWF      HDF      EWF
Min.   :0.0000      Min.   :0.0000      Min.   :0.0000      Min.   :0.0000
1st Qu.:0.0000      1st Qu.:0.0000      1st Qu.:0.0000      1st Qu.:0.0000
Median :0.0000      Median :0.0000      Median :0.0000      Median :0.0000
Mean   :0.0339      Mean   :0.0046      Mean   :0.0115      Mean   :0.0095
3rd Qu.:0.0000      3rd Qu.:0.0000      3rd Qu.:0.0000      3rd Qu.:0.0000
Max.   :1.0000      Max.   :1.0000      Max.   :1.0000      Max.   :1.0000
OSF      RNF
Min.   :0.0000      Min.   :0.0000
1st Qu.:0.0000      1st Qu.:0.0000
Median :0.0000      Median :0.0000
Mean   :0.0098      Mean   :0.0019
3rd Qu.:0.0000      3rd Qu.:0.0000
Max.   :1.0000      Max.   :1.0000
> ai4i_prep <- kNN(ai4i_na)
```



Variable Distributions

Findings:

- Torque follows a normal distribution.
- Rotational speed is positively skewed.
- Other variables do not follow normal distribution.



Correlation Matrix

Association between Type and Machine Failure:

- Performed a Chi-Square test.
- $p\text{-value} = 0.001032 < 0.05$ indicates a statistically significant relationship.

```
> table_quality_failure
      Machine_Failure
Type      0      1
H      982     21
L     5765    235
M     2914     83
```

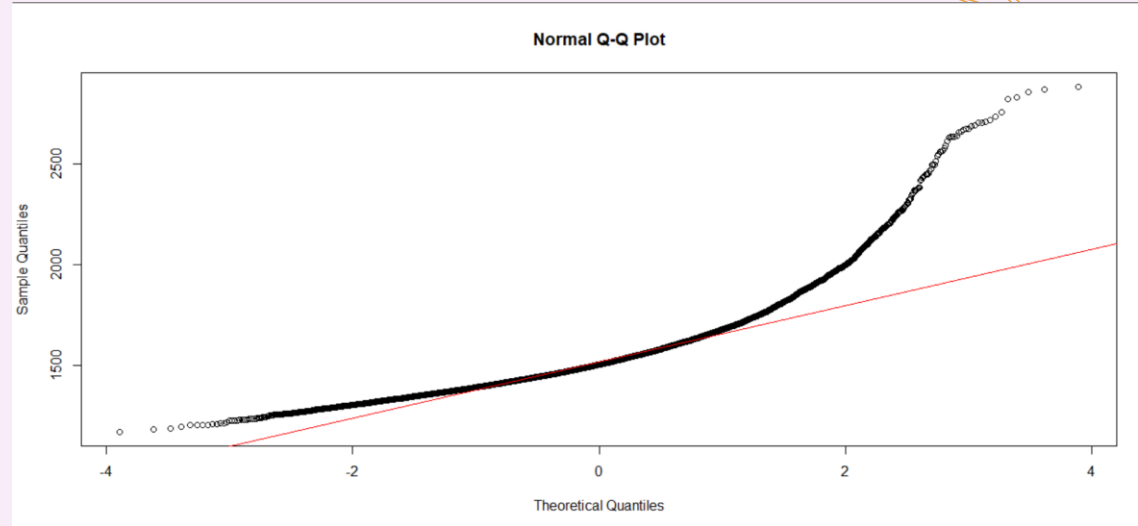
```
> chisq.test(table_quality_failure)
```

```
Pearson's Chi-squared test
```

```
data: table_quality_failure
X-squared = 13.752, df = 2, p-value = 0.001032
```

Normality Tests

- Association between Rotational Speed and Type
 - Normality Tests for Rotational speed: Q-Q Plot for Rotational Speed.
- => The plot showcases deviations in the upper and lower sides suggesting skewness



Kolmogorov-Smirnov Test

Rotational Speed by Type (Low)

p-value < $2.2e-16$ < 5%

Rotational Speed by Type (med)

p-value = $2.206e-10$ < 5%

Rotational Speed by Type (high)

p-value = $2.206e-10$ < 5%

We can conclude that Rotational Speed doesn't follow a normal distribution in any of the categories



Test for intra-group variance

Since normality is not met, we can't use ANOVA we'll use its non-parametric alternative kruskal-wallis since Type has 3 categories (H, L, M)

```
data: Rotational_Speed by Type  
Kruskal-Wallis chi-squared = 0.32541, df = 2, p-value = 0.8498
```

p-value = 0.8498 > 5% meaning there is not a statistically significant difference among the three variant quality products in relation to the rotational speed



Wilcoxon-Mann-Whitney Test for Rotational Speed by Machine Failure

p-value < $2.2e-16$ < 5% : Rotational Speed significantly differs between machines that failed vs. those that did not fail.



Association between Rotational Speed and Tool Wear Failure

ks_twf_no_failure
p-value < 2.2e-16 < 5%

ks_twf_yes_failure
p-value = 0.6761 > 5%

We can conclude that Rotational Speed doesn't follow a normal distribution when there's no TWF but follows a normal distribution when there's TWF



Test for intra-group variance – significant difference

Since normality is met for Rotational Speed during a Tool Wear Failure, let's check homogeneity

p-value = 0.7532 > 5% that means homogeneity is met : variances are equal Thus we can run a T-Test

p-value = 0.7073 : there is no significant difference in Rotational Speed between machines with and without Tool Wear Failure.



Wilcoxon-Mann-Whitney Test for Rotational Speed by Machine Failure

p-value = 0.7059 > 5% : this confirms there is no significant difference in Rotational Speed between machines with and without Tool Wear Failure.

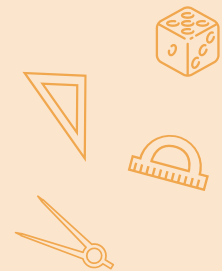


Key Findings

Product Type significantly influences Machine Failure.

Rotational Speed does not follow a normal distribution in any category, and non-parametric tests such as a Kruskal-Wallis and t-test were used.

Rotational Speed significantly differs between machines that failed vs. those that did not fail, but there is no significant difference in Rotational Speed based on Tool Wear Failure.

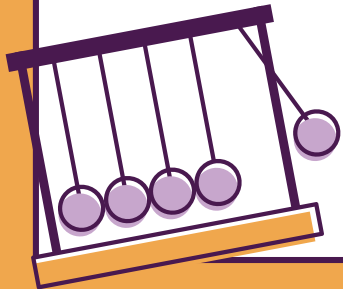


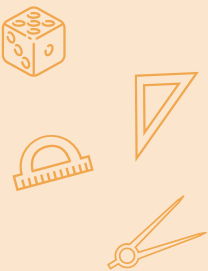


Conclusions

“This is a quote, words full of wisdom
that someone important said and that
can inspire the reader.”

—Someone Famous





9h 55m 23s

Jupiter's rotation period

333,000

The Sun's mass compared to Earth's

386,000 km

Distance between Earth and the Moon



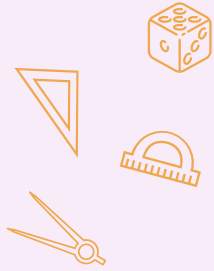


A picture is worth a thousand words



A picture always reinforce the concept

Images reveal large amounts of data, so remember: use an image instead of a long text. Your audience will appreciate it.

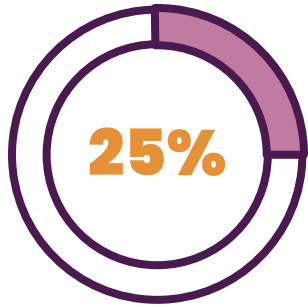




4,498,300,000

Big numbers catch your audience's attention

Let's use some percentages



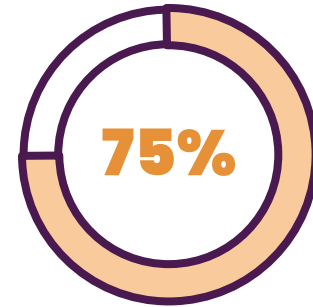
Mercury

Mercury is the closest planet to the Sun and the smallest of them all



Venus

Venus has a beautiful name and is the second planet from the Sun

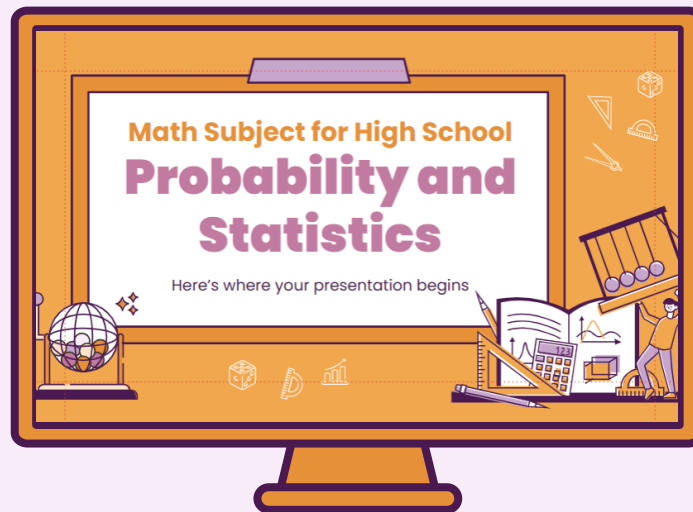


Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

Computer mockup

You can replace the image on the screen with your own work. Just right-click on it and select "Replace image"





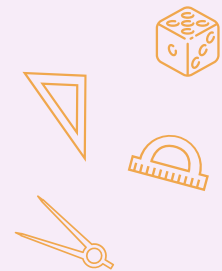
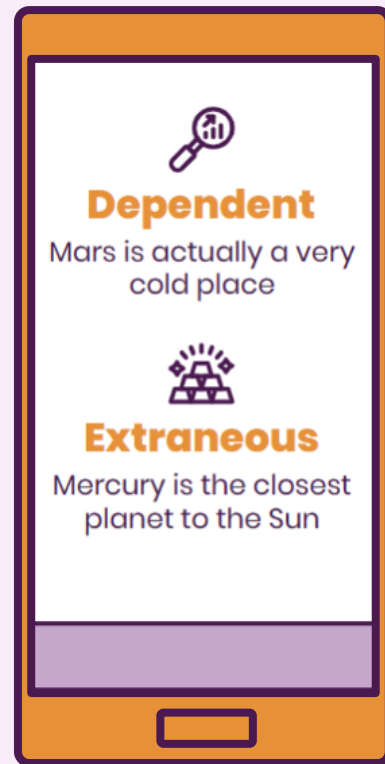
Tablet mockup

You can replace the image on the screen with your own work. Just right-click on it and select "Replace image"



Phone mockup

You can replace the image on the screen with your own work. Just right-click on it and select "Replace image"



This is a map

Venus

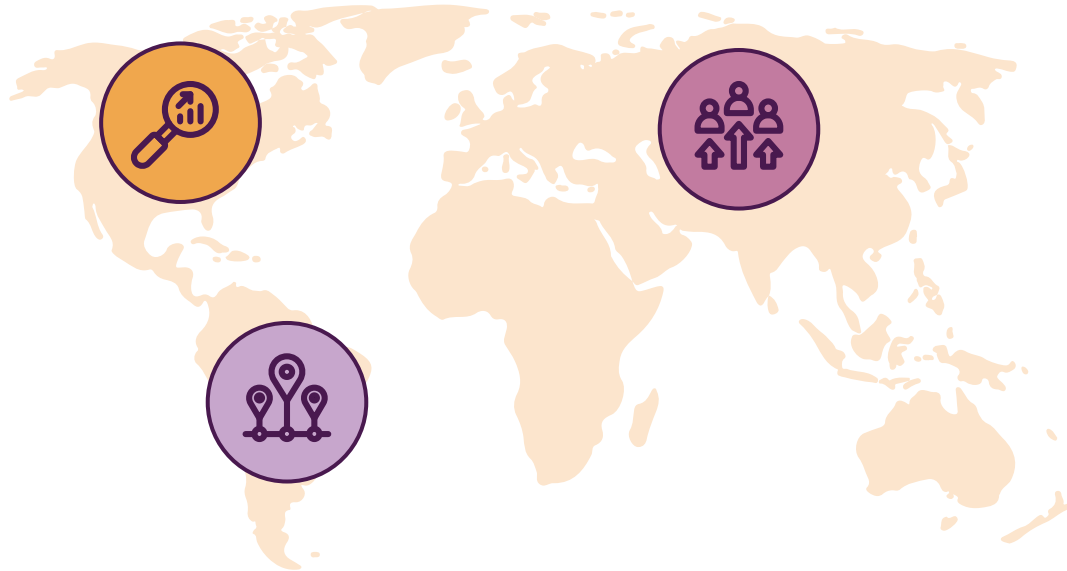
Venus is the second planet from the Sun

Mercury

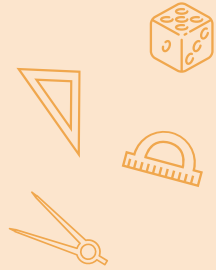
Mercury is the closest planet to the Sun

Mars

Despite being red, Mars is a very cold place



A timeline always works well



Venus is the second planet from the Sun

Venus

Despite being red, Mars is a very cold place

Mars



Mercury

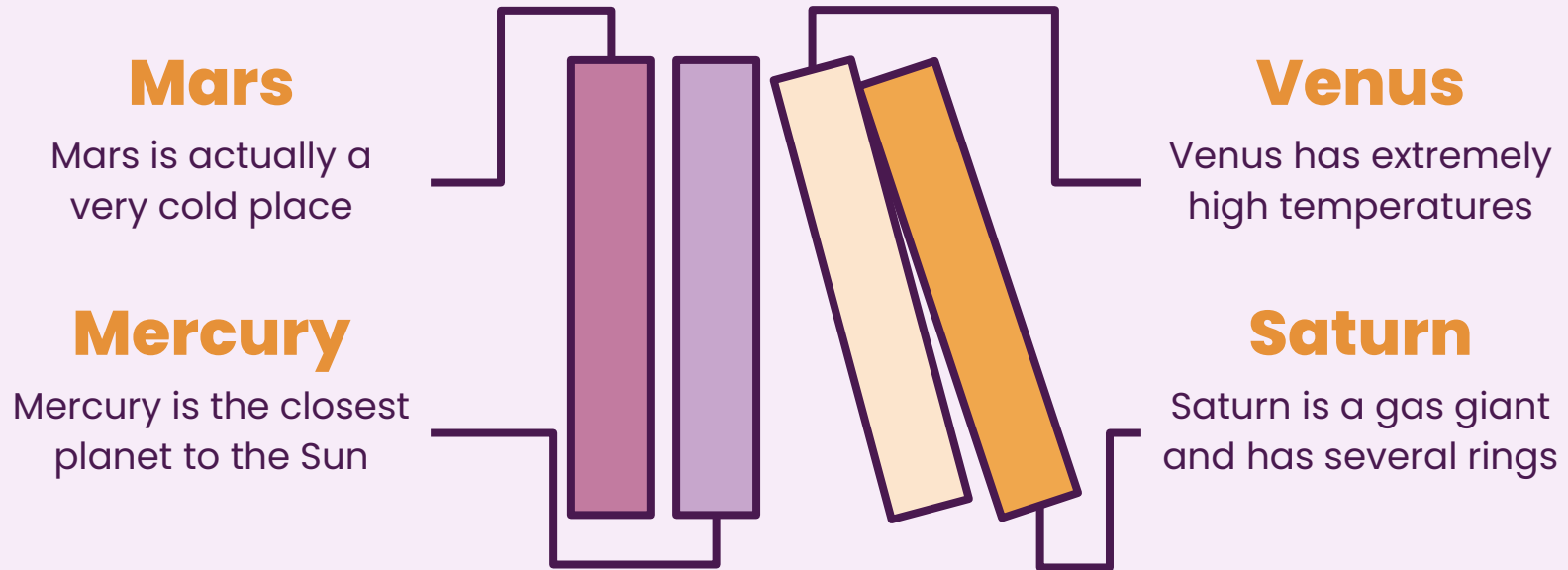
Mercury is the closest planet to the Sun

Jupiter

Jupiter is the biggest planet of them all



Infographics make your idea understandable



Tables represent your data in a clear way

	Mass (earths)	Diameter (earths)	Surface gravity (earths)
Mercury	0.06	0.38	0.38
Mars	0.11	0.53	0.38
Saturn	95.2	9.4	1.16

You can use this graph

Mercury

Mercury is the smallest planet

Jupiter

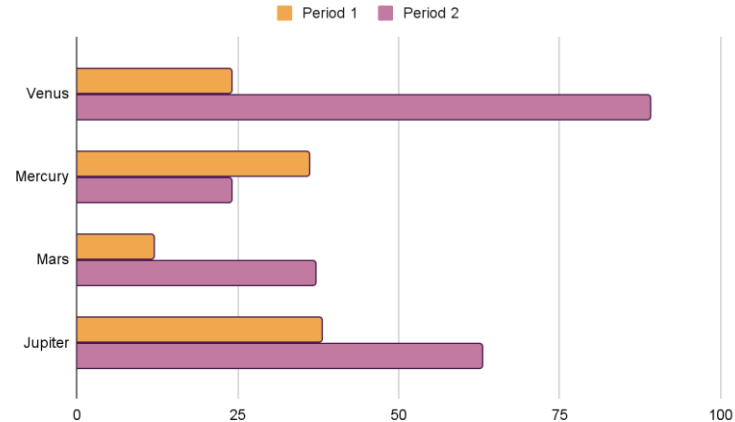
Jupiter is the biggest planet

Venus

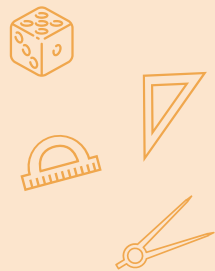
Venus has very high temperatures

Saturn

Saturn is a gas giant with rings



Follow the link in the graph to modify its data and then paste the new one here. For more info, click [here](#)



Our faculty



Sofia Hill

You can speak a bit about this person here

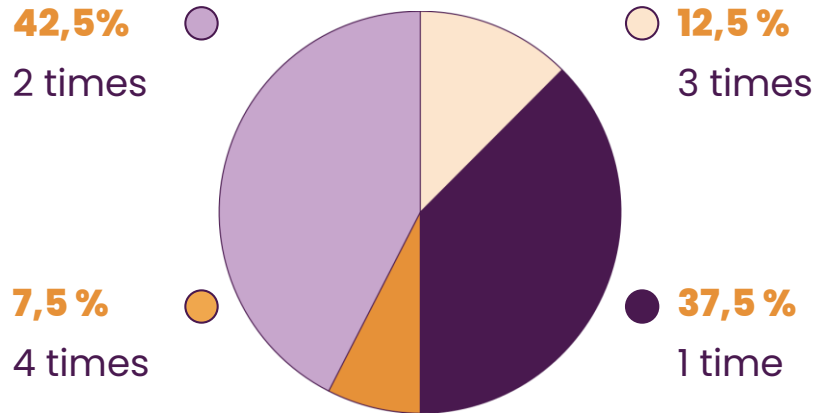


Kaliyah Harris

You can speak a bit about this person here



Exercise 1. Frequency



Follow the link in the graph to modify its data and then paste the new one here. For more info, click **here**

In a class with 40 children in total, they have been asked about the number of times they do sports per week.

Look at the chart and answer. What frequency is repeated the most?





Interpreting two-way tables

	Water	Juice
Boys	8	3
Girls	7	5

In a class, boys and girls have been asked about their food preferences. Look at both tables.

	Salty	Sweet
Boys	5	7
Girls	6	9

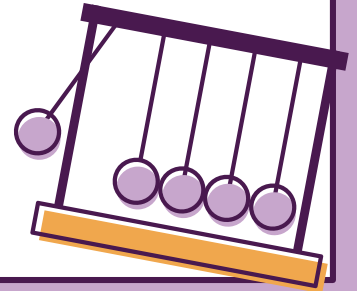
1. How many children prefer to drink water? And juice?
2. How many girls are there in the sweet/salty group?



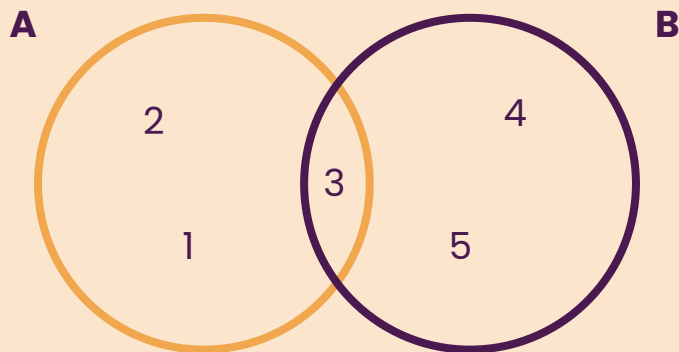
Exercise 2 . Complete the table

	5 meals	4 or 3	TOTAL
Boys	25		40
Girls	30		40

In a class with 40 boys and 40 girls they have been asked about the number of meals they eat per day. **Complete the gap.**



Interpreting two events in probability



☐ $A \cup B =$

☐ ...

☐ ...

☐ ...

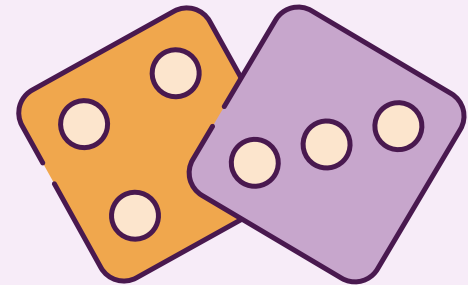
Look at the circles and **name all individual and common events** from cases A and B.



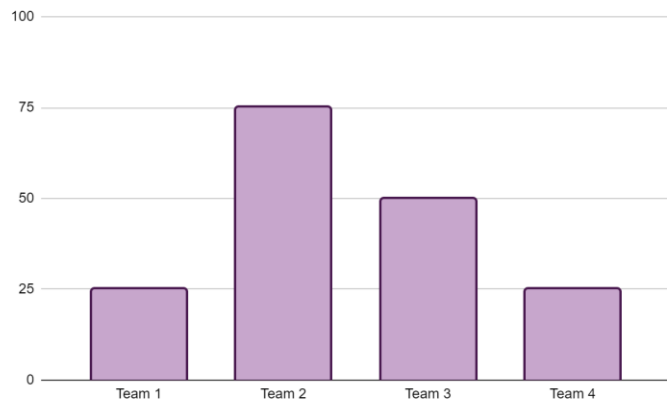
Roll the dice!

Number	1	2	3	4	5	6
Probability	0,12	0,14	0,16	0,17		0,21

By rolling a six-sided die we have obtained the following results for each number. Calculate the probability for the **number 5** and **fill the gap**.



Choose your team!



Follow the link in the graph to modify its data and then paste the new one here. For more info, click **here**

Look at the graphic and **calculate the arithmetic mean and the distribution range** of the score obtained by the following soccer teams.

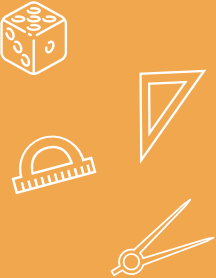


Exercise 3. Events in probability

Sure event	Get a number less than 10
Impossible event	
Possible event	

3	4	5	6
7	5	8	9
2	4	2	1

Pay attention at the numbers and times they appear. Then, complete the table with 3 possible and impossible events for each gap.



Thanks!

Do you have any questions?

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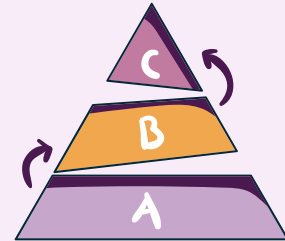
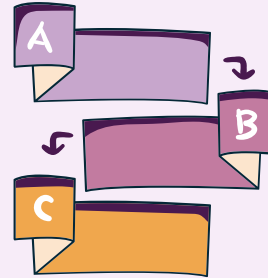
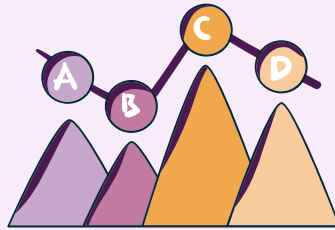
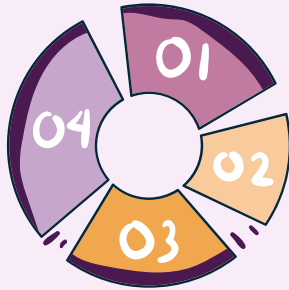


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Here's an assortment of alternative resources whose style fits the one of this template:

Vectors

- Invest data Semi flat illustrations
- Business mission Semi Flat Illustrations
- Professional hand drawn infographic elements





Resources

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- Person doing a presentation for her colleagues
- Man showing some infographics in a video call
- Front view of smiley female teacher posing in front of blackboard
- Female teacher at desk

Icons

- Lottery
- Dice

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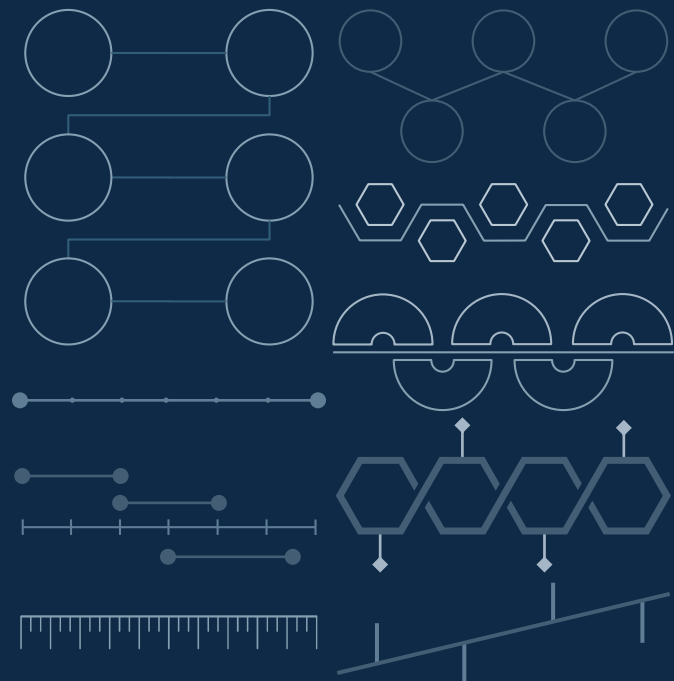
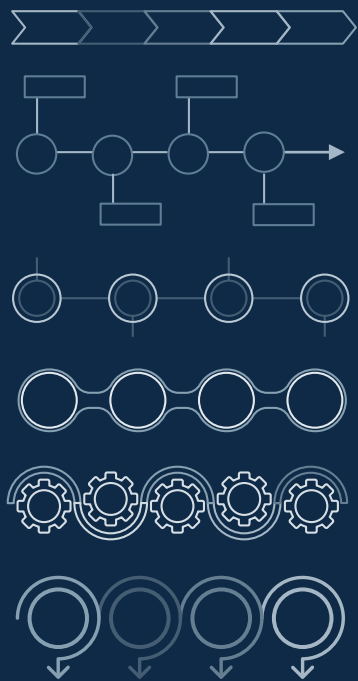
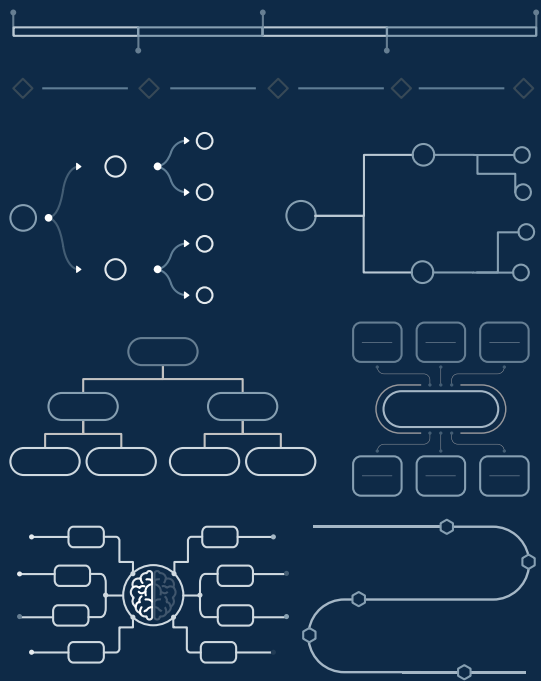
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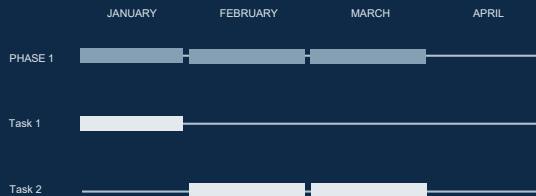
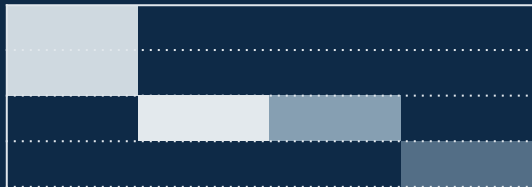
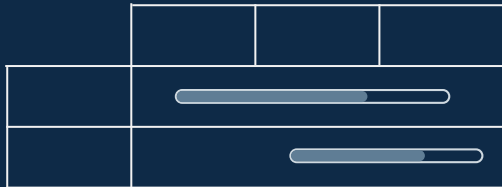
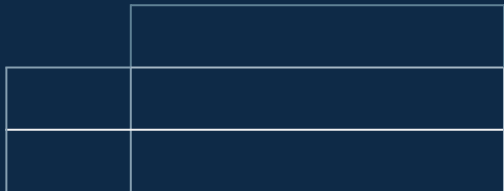
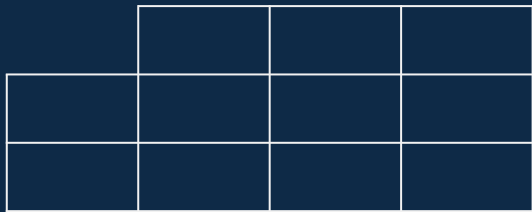
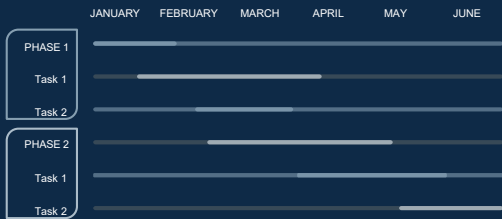
Use our editable graphic resources...

You can easily **resize** these resources without losing quality. To **change the color**, just ungroup the resource and click on the object you want to change. Then, click on the paint bucket and select the color you want. Group the resource again when you're done. You can also look for more **infographics** on [Slidesgo](#).

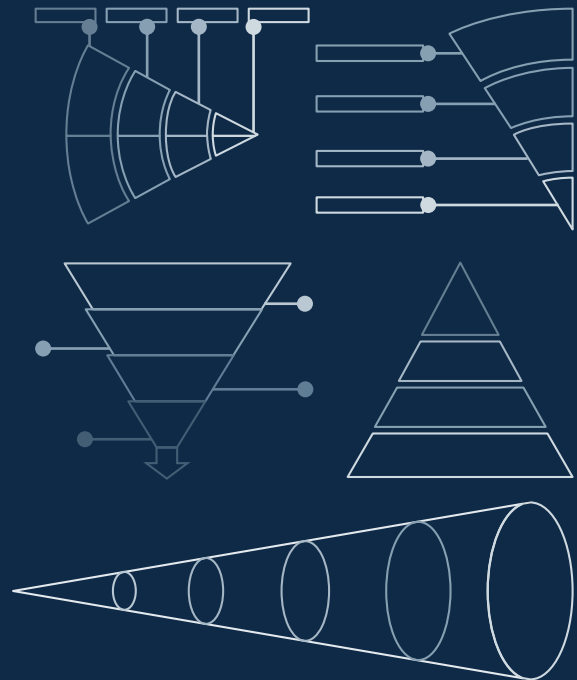
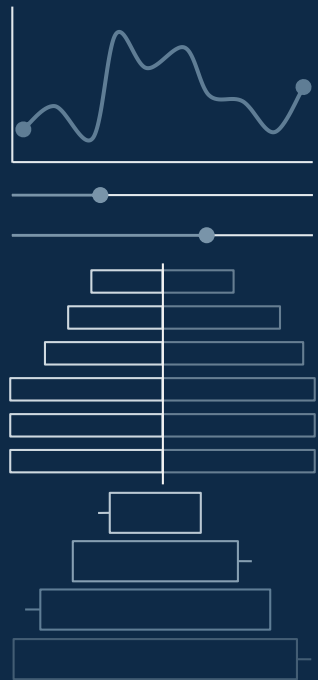
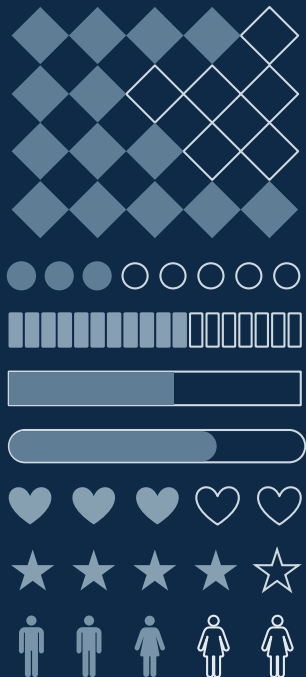
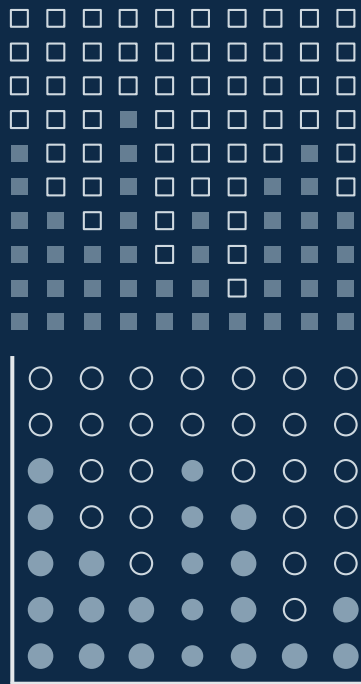












...and our sets of editable icons

You can **resize** these icons without losing quality.

You can **change the stroke and fill color**; just select the icon and click on the **paint bucket/pen**.

In Google Slides, you can also use **Flaticon's extension**, allowing you to customize and add even more icons.



Educational Icons



Medical Icons



Business Icons



Teamwork Icons



Help & Support Icons



Avatar Icons



Creative Process Icons



Performing Arts Icons



Nature Icons



SEO & Marketing Icons



